An Interpretable Credit Scorecard and Approval Cut-Off for Unsecured Personal Loans

Lending Club 36-month loans: development 2012–2014, out-of-time test 2015

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Abstract

A consumer lender needs a transparent way to rank applicants by credit risk and to decide how many of them to approve. Using the Lending Club history of accepted loans, I build a logistic regression scorecard on 36-month loans issued in 2012–2014 ($N = 306{,}462$; bad rate 13.25%) and test it out of time on the 2015 vintage ($N = 283{,}026$; bad rate 14.89%). The model uses only application data and excludes Lending Club's own grade and interest rate.

The scorecard ranks borrowers moderately well and stably: AUC 0.648, Gini 0.296 and KS 0.212 on the 2015 test set, against 0.646, 0.292 and 0.212 in development. Its relative risk ordering is well calibrated (slope 0.996), but it under-predicts the level of default in 2015 by 1.7 percentage points (mean PD 13.2% versus 14.9% observed). FICO score and income are associated with lower default risk; recent credit inquiries, a higher debt-to-income ratio, renting and small-business loans are associated with higher risk.

For policy, the safest 50% of applicants (PD cut-off 0.124) have an observed bad rate of 9.5%, inside a 10% risk appetite, and an expected profit of USD 34.9M. The profit-maximising cut-off depends strongly on the funding cost: it is to approve 90% at 3%, 70% at 5%, and nobody at 7%, where lending is expected to lose money at every cut-off.

1 Introduction

Management of an unsecured personal-loan product has to answer two questions:

1. **Question 1 – Who poses the highest risk of non-payment?** Based on information available at the time of application, can we predict who will default, and how well? This covers (i) which variables best separate good from bad borrowers and how they relate to risk, (ii) the out-of-time discriminatory power (AUC, Gini, KS) and (iii) whether predicted probabilities are calibrated against observed default rates.
2. **Question 2 – Where do we set the approval threshold to avoid losing money?** If we cap risk with a cut-off on the score, what share of applicants do we approve, what is the default rate of the approved portfolio, and which cut-off maximises expected profit? This covers (i) approval rate versus bad rate for percentile cut-offs, compared with the baseline of approving everyone, and (ii) expected profit by cut-off and the sensitivity of the best cut-off to the funding cost.

The answers are written from the estimated coefficients, information values and validation metrics. They describe *associations* with default risk, not causes. The methods follow standard scorecard practice [1, 2, 3].

2 Data and Sample Design

2.1 Source, target and population

The source is the Lending Club accepted-loans file: 2,260,668 loans and 151 columns, of which 43 were kept (31 application variables, 4 benchmark/pricing fields and 8 outcome fields). A loan is **bad** (= 1) if its status is *Charged Off* or *Default* and **good** (= 0) if it is *Fully Paid*. Loans still running (*Current*, *Late*, *In Grace Period*) have no final outcome and are excluded, because treating them as good would understate the bad rate. The *bad rate* is the share of bad loans among loans with a known outcome.

2.2 Vintages, maturity and the out-of-time split

A *vintage* is the group of loans issued in the same calendar year. A 36-month loan issued in December 2015 matures in December 2018, the last month of the file, so all 36-month vintages up to 2015 are essentially complete. Later vintages are right-censored (Table 1, Figure 1).

Table 1: 36-month loans by vintage: volume, share with a final outcome and bad rate.

Vintage	Loans	Resolved	Resolved (%)	Bad rate (%)	Role
2012	43,470	43,470	100.0	13.58	Development
2013	100,422	100,422	100.0	12.33	Development
2014	162,570	162,570	100.0	13.73	Development
2015	283,173	283,026	99.9	14.89	Out-of-time test
2016	323,495	232,361	71.8	19.85	Excluded (censored)
2017	320,419	128,551	40.1	20.02	Excluded (censored)
2018	344,671	41,272	12.0	13.25	Excluded (censored)

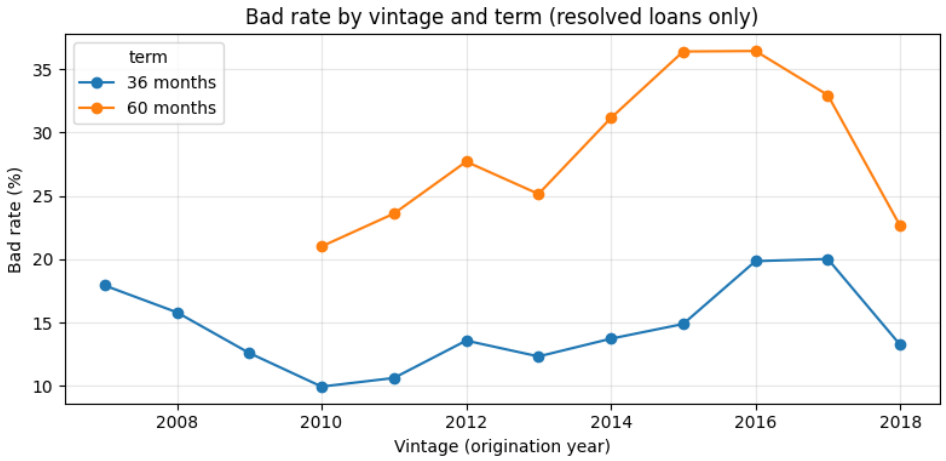


Figure 1: Bad rate by vintage and term (resolved loans only). Vintages from 2016 onward are incomplete (right-censored).

The design that follows from this is:

- **Product:** 36-month loans only. The 60-month product has much higher default (25–36% in the vintages with data) and is still incomplete in 2014–2015.
- **Development set:** vintages 2012–2014, 306,462 loans, baseline bad rate 13.25%.
- **Out-of-time test set:** vintage 2015, 283,026 loans, baseline bad rate 14.89%. Only 147 loans of the 2012–2015 window are still active or late and are excluded.

- Vintages 2007–2011 are left out for low volume and different economic conditions.

Testing on a later period mimics real use, where a model built on the past scores future applicants. A random split would share the same economic conditions between training and test and would overstate performance.

2.3 Variable roles and leakage control

Every kept column was given one role.

- **Application variables** (known at the decision date) are the only model candidates.
- **Outcome variables** (`total_pymnt`, `total_rec_prncp`, `total_rec_int`, `recoveries`, `last_pymnt_d`, `loan_status`) describe what happened *after* the loan was granted. Using them as predictors would be data leakage.
- **Lending Club's own rating and pricing** (`grade`, `sub_grade`, `int_rate`) already embed the lender's own risk assessment, so using them would make the scorecard circular. `int_rate` and `installment` are used only to compute interest income in Question 2.
- **Excluded a priori:** `addr_state` (a geographic proxy, a fair-lending concern); `fico_range_high` (identical to `fico_range_low`, Spearman $\rho = 1$); `term` and `application_type` (constant in the development set).

2.4 Missing values

A variable is eligible only if it has at most 5% missing values in *every* vintage from 2012 to 2015. Otherwise median imputation would turn the variable into a detector of the vintage rather than a measure of risk. Eleven candidates failed this rule (Table 2); sixteen remained.

Table 2: Candidates excluded by the 5% missing-value rule (worst vintage).

Variable	Missing (%)	Worst vintage and reason
`mths_since_last_record`	97.3	2012; empty when no record exists (structural)
`mths_since_last_delinq`	59.1	2012; empty when never delinquent (structural)
`tot_cur_bal`, `total_rev_hi_lim`, `num_tl_op_past_12m`, `pct_tl_nvr_dlq`	51.9	2012; bureau fields introduced during 2012
`bc_util`, `bc_open_to_buy`	14.9, 14.8	2012; same bureau roll-out
`mort_acc`, `acc_open_past_24mths`	13.9	2012; same bureau roll-out
`emp_length_years`	6.7	2015; rises each year from 4.0% in 2012

3 Model Development

All selection steps use the development set only. The test set is never used to choose variables.

3.1 Information value

For each variable the development set is split into ten quantile bins (categorical variables use their own levels). In each bin the *weight of evidence* is

$$\mathrm{WoE}_j = \ln \frac{\%\mathrm{good}_j}{\%\mathrm{bad}_j}, \qquad \mathrm{IV} = \sum_j (\%\mathrm{good}_j - \%\mathrm{bad}_j)\,\mathrm{WoE}_j.$$

A positive WoE marks a bin that is safer than average. By convention IV < 0.02 is useless, 0.02–0.10 weak, 0.10–0.30 medium, 0.30–0.50 strong, and > 0.50 suspicious (a possible leak). Table 3 gives the ranking. Seven variables reach the 0.02 threshold and none is suspicious.

Table 3: Information value of the 16 eligible variables (development set).

Rank	Variable	Type	IV	Strength
1	`fico_range_low`	Numeric	0.1320	Medium
2	`annual_inc`	Numeric	0.0870	Weak
3	`dti`	Numeric	0.0475	Weak
4	`home_ownership`	Categorical	0.0390	Weak
5	`inq_last_6mths`	Numeric	0.0286	Weak
6	`purpose`	Categorical	0.0226	Weak
7	`revol_util`	Numeric	0.0207	Weak
8	`credit_history_years`	Numeric	0.0199	Useless
9	`revol_bal`	Numeric	0.0178	Useless
10	`total_acc`	Numeric	0.0093	Useless
11	`verification_status`	Categorical	0.0092	Useless
12	`loan_amnt`	Numeric	0.0078	Useless
13	`pub_rec_bankruptcies`	Numeric	0.0019	Useless
14	`delinq_2yrs`	Numeric	0.0011	Useless
15	`pub_rec`	Numeric	0.0005	Useless
16	`open_acc`	Numeric	0.0003	Useless

3.2 Redundancy and multicollinearity

Two variables are redundant if their Spearman rank correlation is at least 0.70 in absolute value. None of the seven selected variables was dropped on this rule. The variance inflation factor (VIF) measures how much a coefficient's variance is inflated by correlation with the other predictors. The final VIFs for the numeric variables are `revol_util` 1.38, `fico_range_low` 1.35, `annual_inc` 1.13, `dti` 1.11 and `inq_last_6mths` 1.04, far below the ceiling of 5.

3.3 Preprocessing and estimation

Preprocessing parameters are learned on the development set and applied unchanged to the test set, which avoids leakage. Numeric variables are winsorised at the 1st and 99th percentiles, log-transformed if non-negative and strongly right-skewed (skewness > 1), median-imputed and standardised. Categories below 1% of the data are pooled into a `RARE` level, and dummies are created against the most frequent category (mortgage for housing, debt consolidation for purpose).

The scorecard is a logistic regression estimated by maximum likelihood, with no penalty and no class weights, so that the predicted probabilities of default (PD) are honest:

$$\ln \frac{p_i}{1 - p_i} = \beta_0 + \sum_k \beta_k x_{ik}.$$

The fit converged. With about 40,600 defaults and 17 parameters, the events per parameter are 2,388, far above the usual minimum of 10. Because numeric inputs are standardised, an odds ratio is the multiplicative change in the odds of default for a +1 standard deviation in the variable (for dummies, versus the reference category). Table 4 reports the estimates.

Table 4: Logistic regression, development set 2012–2014 ($N = 306{,}462$).

Parameter	Coef.	Std. err.	z	p	Odds ratio	95% CI
Intercept	−2.0738	0.0099	−209.26	< 0.001	0.126	[0.123, 0.128]
`fico_range_low`	−0.3214	0.0072	−44.89	< 0.001	0.725	[0.715, 0.735]
`annual_inc`	−0.2099	0.0061	−34.34	< 0.001	0.811	[0.801, 0.820]
`inq_last_6mths`	+0.1668	0.0054	+30.90	< 0.001	1.182	[1.169, 1.194]
`dti`	+0.1566	0.0056	+27.78	< 0.001	1.170	[1.157, 1.183]
`revol_util`	+0.0425	0.0063	+6.72	< 0.001	1.043	[1.031, 1.057]
home_ownership = RENT	+0.2423	0.0122	+19.92	< 0.001	1.274	[1.244, 1.305]
home_ownership = OWN	+0.1248	0.0196	+6.38	< 0.001	1.133	[1.090, 1.177]
home_ownership = RARE	+0.3994	0.3075	+1.30	0.194	1.491	[0.816, 2.724]
purpose = small_business	+0.7757	0.0412	+18.84	< 0.001	2.172	[2.004, 2.355]
purpose = credit_card	−0.2152	0.0136	−15.77	< 0.001	0.806	[0.785, 0.828]
purpose = medical	+0.2497	0.0485	+5.14	< 0.001	1.284	[1.167, 1.412]
purpose = other	+0.1419	0.0235	+6.03	< 0.001	1.153	[1.101, 1.207]
purpose = RARE	+0.1725	0.0347	+4.97	< 0.001	1.188	[1.110, 1.272]
purpose = car	−0.1646	0.0590	−2.79	0.005	0.848	[0.756, 0.952]
purpose = home_improvement	+0.0413	0.0263	+1.57	0.116	1.042	[0.990, 1.097]
purpose = major_purchase	−0.0076	0.0419	−0.18	0.856	0.992	[0.914, 1.077]

Linearity check. The model is linear in the log-odds, so each numeric variable should show a monotonic pattern of bad rate across its deciles. This holds for all five (Spearman between decile order and bad rate): `fico_range_low` −1.00 (bad rate from 18.0% in the lowest decile to 5.6% in the highest), `annual_inc` −1.00 (19.6% → 8.5%), `dti` +1.00 (10.1% → 18.1%), `inq_last_6mths` +1.00 (12.4% → 17.4%) and `revol_util` +0.99 (9.9% → 15.9%).

4 Question 1: Who Poses the Highest Risk of Non-Payment?

4.1 Which variables separate good from bad borrowers, and how do they relate to risk?

By information value, the FICO score is the most informative variable (IV 0.132), followed by income (0.087) and debt-to-income ratio (0.048). All others are weak. Table 4 shows the direction and size of each association, holding the other variables fixed.

- **Associated with lower default risk:** a higher FICO score (odds ratio 0.73 per +1 SD, i.e. 27.5% lower odds) and a higher income (0.81, 18.9% lower odds). Loans taken to pay off credit-card debt have 19.4% lower odds than debt-consolidation loans (0.81), and car loans 15.2% lower (0.85).
- **Associated with higher default risk:** more credit inquiries in the last six months (1.18), a higher debt-to-income ratio (1.17), renting rather than having a mortgage (1.27), medical loans (1.28) and above all small-business loans, whose odds are more than double (2.17). Revolving utilisation has a small positive association (1.04).
- **No reliable association:** `home_ownership` = RARE, `purpose` = home_improvement and `purpose` = major_purchase have $p > 0.05$. Their coefficients should not be interpreted.

With 306,462 observations almost every p-value is tiny, so the size of the odds ratio and its confidence interval matter more than significance. The 27.5% and 18.9% effects of FICO and income are the largest in practical terms. These are associations in an approved population, not causal effects.

4.2 Discriminatory power out of time

Table 5: Discrimination and calibration-in-the-large, development versus out-of-time test.

Sample	AUC	Gini	KS	Observed bad rate	Mean PD
Development (2012–2014)	0.6457	0.2915	0.2115	13.25%	13.25%
Test (2015)	0.6479	0.2958	0.2116	14.89%	13.16%

The AUC is the probability that a randomly chosen bad loan receives a higher PD than a randomly chosen good loan (0.5 is a coin flip), the Gini coefficient is $2\,\mathrm{AUC} - 1$, and the KS statistic is the largest gap between the cumulative PD distributions of bad and good loans. Out of time the model reaches AUC 0.648, Gini 0.296 and KS 0.212 (Table 5). Performance is the same as in development, so there is no sign of overfitting or of a loss of ranking power in 2015. The level is modest: Gini around 0.30 is at the lower end of what is usually seen in consumer credit. This is expected on a population that Lending Club has already screened, and with only seven weak-to-medium predictors.

4.3 Calibration

Calibration asks whether predicted probabilities match observed default rates. On the 2015 test set:

- **Calibration slope** = 0.996 (ideal 1): the spread of the predicted risk across borrowers is correct.
- **Calibration intercept** = +0.141 (ideal 0): observed default is higher than predicted. Mean PD is 13.16% against an observed 14.89%, a gap of 1.73 percentage points.

Table 6 and Figure 2 show the same result by PD decile. Risk rises strictly from 4.43% in the safest decile to 27.31% in the riskiest, a 6.2-fold difference. The safest decile is almost exact, and from the second decile on the observed rate is above the prediction by roughly 0.7 to 2.5 points.

Table 6: Calibration by predicted-PD decile, test vintage 2015.

Decile	Loans	Mean PD	Observed bad rate	Gap (pp)
1 (safest)	28,303	4.41%	4.43%	+0.02
2	28,303	6.93%	7.62%	+0.69
3	28,302	8.65%	10.09%	+1.44
4	28,303	10.17%	11.84%	+1.67
5	28,302	11.64%	13.74%	+2.10
6	28,303	13.17%	15.31%	+2.14
7	28,302	14.84%	17.39%	+2.55
8	28,303	16.84%	19.20%	+2.36
9	28,302	19.54%	21.93%	+2.39
10 (riskiest)	28,303	25.35%	27.31%	+1.96

The pattern (correct ranking and slope, higher level) means that, for the same applicant profile, default was more frequent in 2015 than in 2012–2014. The average predicted PD of 2015 applicants

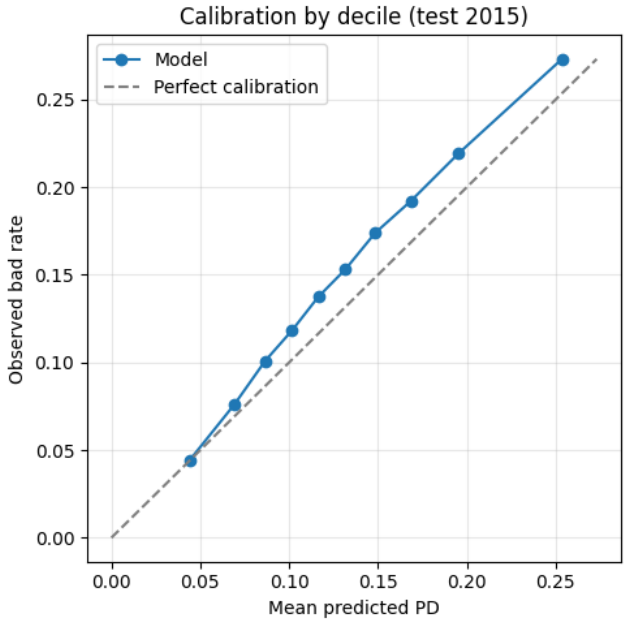


Figure 2: Calibration by PD decile on the 2015 test set. The curve is parallel to the diagonal and shifted upwards.

(13.16%) is close to the development average (13.25%), so the shift is in the default behaviour of comparable profiles rather than in the mix of applicants. This study does not test why. Candidates are the economic environment, changes in lending practice and information that the seven variables do not capture.

4.4 Answer to Question 1

Yes, default can be predicted from application data, but only moderately. FICO score, income, debt-to-income ratio, recent credit inquiries, housing status and loan purpose are the variables associated with default risk, with FICO and income the most informative. Out of time the scorecard has AUC 0.648, Gini 0.296 and KS 0.212, unchanged from development. Its risk ordering is well calibrated, but its PDs are about 1.7 percentage points too low in 2015, so they should not be read as absolute default rates without a level adjustment.

5 Question 2: Where Do We Set the Approval Threshold?

5.1 Approval rate versus bad rate

The cut-off is defined on percentiles of the PD ranking: approving q% means accepting the q% of 2015 applicants with the lowest PD. The *baseline* is the bad rate when everybody is approved, 14.89%. Table 7 and Figure 3 show how the bad rate of the approved portfolio rises with the approval rate.

Every approval rate below 100% beats the baseline. Approving only the safest 10% gives a bad rate of 4.4%, about one third of the baseline, and approving 50% gives 9.5%, about two thirds. Against a management risk appetite of 10% (an assumption, not an estimate), the largest acceptable approval rate is 50%, with a PD cut-off of 0.124; at 55% the bad rate is already 10.03%.

Table 7: Approval rate versus bad rate of the approved portfolio, test vintage 2015.

Approval	Approved	PD cut-off	Mean PD	Bad rate	vs. baseline
10%	28,303	0.0593	4.41%	4.43%	0.30
15%	42,454	0.0696	5.09%	5.30%	0.36
20%	56,605	0.0784	5.67%	6.03%	0.40
25%	70,756	0.0866	6.19%	6.73%	0.45
30%	84,908	0.0943	6.67%	7.38%	0.50
35%	99,059	0.1018	7.11%	7.97%	0.54
40%	113,210	0.1090	7.54%	8.49%	0.57
45%	127,362	0.1164	7.96%	9.00%	0.60
50%	**141,513**	**0.1238**	**8.36%**	**9.54%**	**0.64**
55%	155,664	0.1316	8.76%	10.03%	0.67
60%	169,816	0.1397	9.16%	10.50%	0.71
65%	183,967	0.1482	9.57%	11.01%	0.74
70%	198,118	0.1578	9.97%	11.49%	0.77
75%	212,270	0.1682	10.39%	11.94%	0.80
80%	226,421	0.1801	10.83%	12.45%	0.84
85%	240,572	0.1948	11.30%	12.95%	0.87
90%	254,723	0.2138	11.80%	13.51%	0.91
95%	268,875	0.2435	12.37%	14.16%	0.95
100%	283,026	0.5591	13.16%	14.89%	1.00

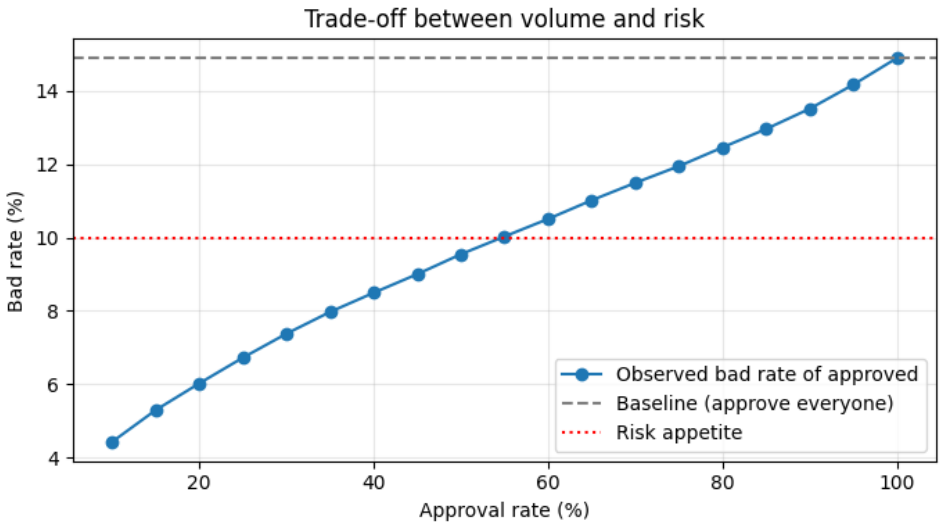


Figure 3: Trade-off between approval volume and risk on the 2015 test set.

5.2 Expected profit

Expected profit per loan is computed before the outcome is known:

$$\mathbb{E}[\Pi_i] = (1 - \mathrm{PD}_i)\, I_i - \mathrm{PD}_i \cdot \mathrm{LGD} \cdot \mathrm{EAD}_i - c \cdot \mathrm{EAD}_i \cdot T,$$

where EAD_i is the loan amount, $I_i = 36 \times \text{installment}_i - \text{loan amount}_i$ is the contractual interest if all instalments are paid, $\mathrm{LGD} = 65\%$ is the loss given default (a conservative value in the usual 60–70% range for unsecured credit), c is the annual funding cost and $T = 1.5$ years is the average outstanding life of a 3-year amortising loan. The base funding cost is $c = 5\%$. The expected profit of a cut-off is the sum over the approved loans. The grid includes 0% approval (profit 0), so that the optimiser can reject everyone when lending loses money.

5.3 Sensitivity of the best cut-off to the funding cost

Table 8: Expected profit (USD million) by approval rate and funding cost, test vintage 2015.

Approval rate	Funding 3%	Funding 5% (base)	Funding 7%
0%	0.00	0.00	0.00
10%	22.56	7.48	−7.60
20%	44.63	15.62	−13.39
30%	65.26	23.41	−18.44
40%	83.59	29.86	−23.86
50%	99.67	**34.89**	−29.88
60%	112.77	37.68	−37.41
70%	123.50	**38.77**	−45.96
80%	130.88	37.20	−56.49
90%	**133.88**	32.06	−69.75
100%	128.73	20.01	−88.72



Figure 4: Expected profit by approval rate under three funding costs (test vintage 2015).

With the base funding cost the profit curve is flat near its top: expected profit is USD 38.8M at 70% approval, 38.6M at 65% and 37.7M at 60% (Table 8). The unconstrained optimum (70%) has a bad rate of 11.5%, above the 10% risk appetite. Respecting the appetite means approving

Table 9: Profit-maximising approval rate by funding cost.

Funding	Best approval	PD cut-off	Bad rate	Exp. profit (USD M)
3%	90%	0.2138	13.51%	133.88
5% (base)	70%	0.1578	11.49%	38.77
7%	0%	—	—	0.00 (none profitable)

50% of applicants, at a cost of about USD 3.9M of expected profit relative to the unconstrained optimum.

The best cut-off is very sensitive to the funding cost (Table 9). At 3% almost everybody is worth approving (90%), whereas at 7% expected profit is negative at every positive approval rate, and the best decision is to approve nobody. Funding cost is therefore the assumption that matters most for the policy.

5.4 Answer to Question 2

At the base assumptions, the policy that respects a 10% bad-rate appetite approves the safest 50% of applicants (PD cut-off 0.124). The approved portfolio has a bad rate of 9.5%, against 14.9% if everybody were approved, and an expected profit of USD 34.9M. Without the risk-appetite limit, expected profit peaks at 70% approval (USD 38.8M) but with an 11.5% bad rate. This recommendation holds only for funding costs around 5%: at 3% the optimum moves to 90% approval and at 7% lending is not profitable at any cut-off.

6 Limitations

1. **Selection bias.** Only loans that Lending Club approved are observed. Results apply to that population, and the behaviour of rejected applicants is unknown (reject inference is not attempted).
2. **Expected profit is indicative.** It assumes full contractual interest, a constant 65% loss on the original amount, a flat funding cost over a fixed 1.5-year life and no operating costs. It has not been compared with the cash actually received, so profit levels should be read with caution and the cut-off as sensitive to these assumptions.
3. **PDs are too low in 2015.** Because expected loss uses the model's PD, which under-predicts default by about 1.7 points, the expected profits above are likely optimistic.
4. **Cut-offs are chosen and evaluated on the same test vintage.** In production the cut-off would be fixed on past data before applying it to new applicants.
5. **Variables lost to data availability.** Requiring $\leq 5\%$ missing values in 2012 removed several bureau variables that are complete from 2013 onward (Table 2). A development window starting in 2013 could recover them.
6. **One credit cycle.** 2012–2015 was a benign, expanding period, and the PDs may not hold in a downturn.
7. **Independence.** A borrower may hold more than one loan, which cannot be checked here.
8. **Scope.** Only the 36-month product is covered, and coefficients describe who tends to default, not why.

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